

NZLUM v0.4 & the 2026 land use map

Changes, validation & recommendations

NZ Land Use Steering Group (NZLUSG) – Meeting 2

Richard Law & James Ardo, BSI · 2026



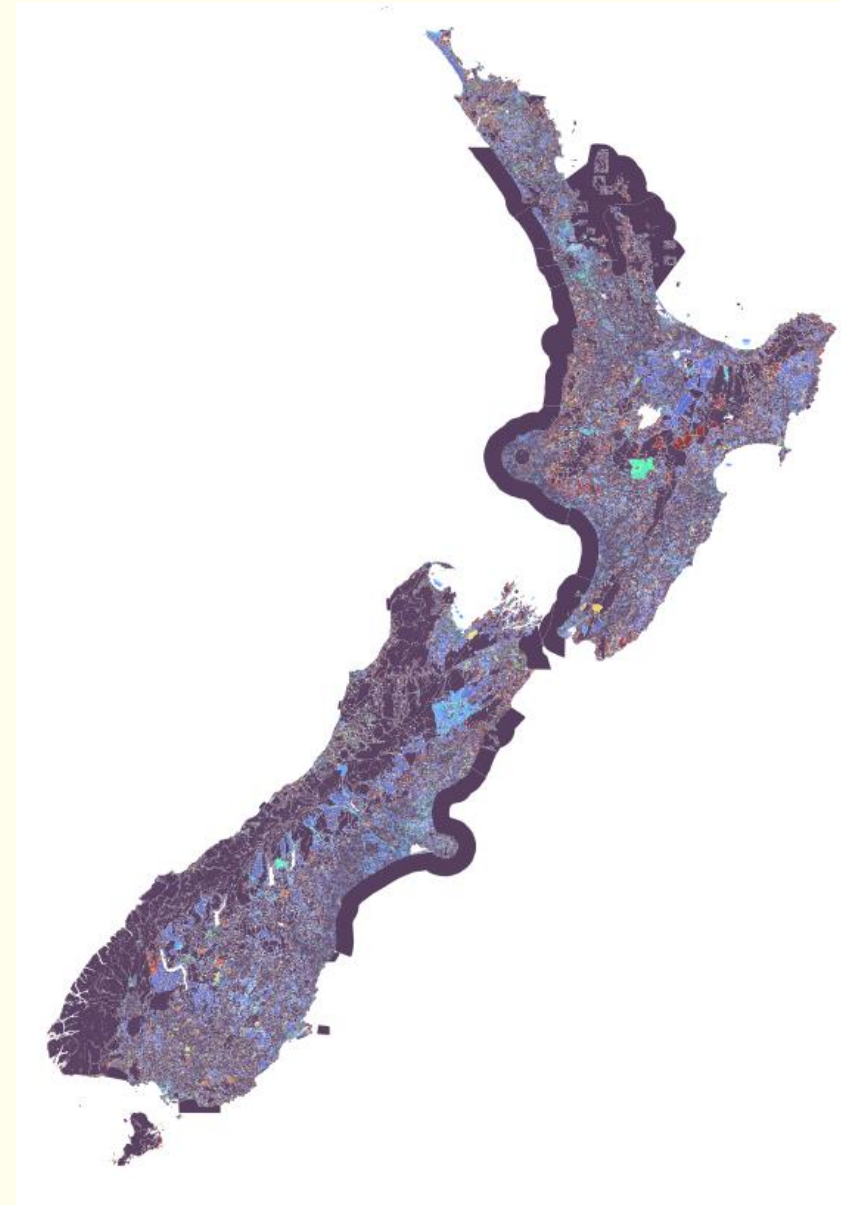
Introduction and pointers



- **Published:** NZLUM v0.4 2026 national land use map (July 2026), implementing the revised v0.3 → v0.4 classification
- **This meeting:** review v0.4 and the 2026 map, then decide what further change is worth prioritising before MCERT and BSI commit
- Expectation over time: progressively **less change to the classification system**, more attention to **validation performance and spatial precision** of the map
- Still open to ideas; we want a sense of agreement before proceeding
- **Everything referenced today is at <https://nzlum.landcareresearch.co.nz>**
 - Classification v0.4 with change notes, summary PDF, data structure
 - Governance: NZLUSG terms of reference, Meeting 1 agenda and minutes
 - Documents: LC4620 (2025 method) and the 2026 addendum report 2627-0013
 - Data: LRIS set 7516 “NZLUM v0.4 2026”
 - Classifier configuration on GitHub ([manaakiwhenua/luis-config-nzlum](https://github.com/manaakiwhenua/luis-config-nzlum))

Recap: a specification and a map

- **NZLUM v0.4** = the classification system: classes, definitions, hierarchy, attributes, schema (SKOS vocabulary)
- **NZLUM v0.4 2026** = one implementation: a national map from 117 input layers, on the H3 discrete global grid
- **Classifier:** a rule-based expert system with competitive evidence scoring (a spatial knowledge-based scoring classifier)
 - Each class starts from a base confidence set by its strongest evidence, then adds bonuses/penalties from corroborating or contradicting layers
 - Hard exclusions remove impossible classes; lowest (most confident) score wins; ties broken by recency and geographic precision
 - Analogous to Dempster–Shafer accumulation but with expert ordinal weights, not calibrated probabilities: no independent ground truth exists to calibrate against
- Deterministic, transparent, auditable; limited by how well rules encode knowledge and by input data quality, recency, precision and completeness



Part 1 - What NZLUSG decided in January 2026



Classification system (→ v0.4)

- **2.3.0** Short-rotation cropping: explicit 5-year periodicity for rotational crops
- **2.2.2** Intensive dry stock: removed sentence sending deer, goats and horses to extensive class
- **2.6.4** Horse studs: intensive parts only; extensive parts → 2.2.3
- **New 3.1.4** Rural residential (0.4–2 ha, dwelling, zoning) and **new 2.9.0** Smallholder farm (2–40 ha, residential motivation)
- **3.6.0** renamed Transportation; navigation & communication moved to utilities as **3.5.8** (was 3.6.5)
- No new attributes; permeability stays optional

2026 mapping practice

- Priority: map **2.2.0** grazing modified pasture at tertiary level (2.2.1 dairy, 2.2.2 intensive, 2.2.3 extensive dry stock)
- Elaborate **Class 3** without new data acquisition; BSI to choose which tertiary classes
- → BSI mapped 3.1.1–3.1.4 residential, 3.2.1–3.2.3 recreation & community services, plus 2.9.0
- Draft to NZLUSG in May; release June/July; Meeting 2 to review validation
- All decisions by consensus; 16-item actions & decisions register (A–P) in the minutes

Peri-urban space: 3.1.4 and 2.9.0



3.1.4 Rural residential (Class 3)

- Residential in rural/peri-urban setting; land management incidental to living there (≈ Rural lifestyle zone)
- **Size:** 0.4–2 ha with a dwelling; > 2 ha → Class 2 even if NDVR says residential/lifestyle; 0.4–2 ha without dwelling = in transition
- **Evidence:** NDVR actual-use 21/22 (lifestyle) with confirmed dwelling, 97 (bach, rural), category L (LB/LI/LV)
- **Ruled out by:** > 2 ha, tiny urban parcels, high-density indicators, industrial zoning

2.9.0 Smallholder farm (Class 2, new)

- Rural property > 2 ha where the primary motivation is residential, not commercial production; management low-intensity
- Proposed by Haydon Jones (WRC); thresholds from land-fragmentation size classes (2 / 40 ha)
- **Evidence:** NDVR > 2 ha with actual-use 91, 21 or 22, or (lower confidence) property code LI; improvements text implies a dwelling (not townhouses/apartments)
- Zoning bonus/penalty; confidence slides down toward 40 ha (< 10 ha strongest); excluded by winter forage crops or incompatible LCDB cover
- Meeting 1 decision D: tertiary size gradation “at a future point” → **for today?**

Other v0.4 changes (agreed internally at BSI before NZLUSG)

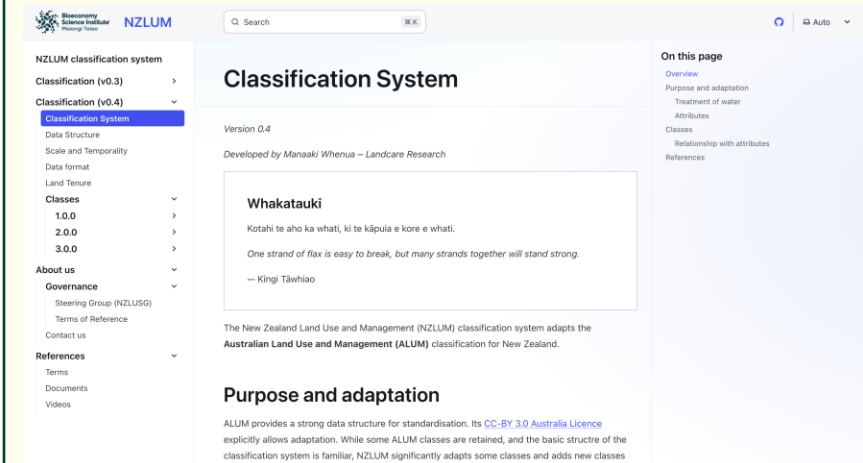


- **2.1.1** Exotic plantation forestry: explicitly includes plantations for carbon sequestration / credits, not only physical commodities
- **1.3.4** Production from indigenous vegetation: clarified as non-plantation native forest
- **1.4.0 / 2.8.0 / 3.9.0** “Unused/Vacant and transitioning land”: labels made consistent across the three primary classes
 - Their subclasses re-ordered by descending likelihood of leaving the category, e.g. **3.9.1** Brownfield development, **3.9.2** Greenfield development, **3.9.3** Idle and derelict land
- **3.6.0** Transportation: navigation/communication removed; **3.5.8** Navigation and communication created under Utilities (was 3.6.5)
- Every change is recorded as a SKOS change note, shown under the class definition on the website
- v0.3 remains published (deprecated) so the 2025 map stays interpretable

Publication: a versioned SKOS vocabulary

- v0.3 and v0.4 both published as **SKOS concept schemes**, validated with **SHACL** (including the hierarchy)
- Canonical form is Turtle; the website pages are generated from it; XLSX and Markdown are derived re-presentations; more formats on request
- Change notes, scope notes and examples attached to classes; also controlled lists for commodities and management practices; possible tenure, zoning and water in the future.
- **Semantic versioning:** v0.4.1 = compatible clarification; v0.5 = departures that affect the map
- Intent: changes become progressively minor to avoid disrupting end users and to keep the 2025/2026/... maps comparable

<https://nzlum.landcareresearch.co.nz/classification/v0-4/system>



RDF (SKOS + SHACL)



RDF – formal, structured,
authoritative, single-source of truth

```
nzsluc / classification-systems / nzlum / rdf / 0.4.0 / concepts-2.ttl
Code Blame 444 lines (402 loc) · 26.4 KB
103 luc:colour "#357735" ;
104 skos:prefLabel "Dairy"@en ;
105 skos:definition "Land on which milking cows (or other stock, such as goats or sheep) are grazed during the milking" ;
106 skos:scopeNote "Dairy production systems can include rotations of grazed forage crops and maize for silage, and dry-stock grazing, but this class should only be used where dairy is the primary purpose of the land." ;
107 skos:scopeNote "Where the land is permanently used for dry-stock grazing, it should be classified under dry-stock grazing." ;
108 .
109
110 luc:intensive_dry_stock a skos:Concept ;
111 skos:inScheme lus:nzslum0.4.0 ;
112 skos:broader luc:grazing_modified_pasture_systems ;
113 skos:notation "2.2.2" ;
114 lu:colour "#357735" ;
115 skos:prefLabel "Intensive dry stock"@en ;
116 skos:definition "Non-milking platform grazing land with higher inputs (e.g. fertiliser), higher likelihood of irrigation, and higher stocking rates; typically on flat or rolling terrain." ;
117 skos:scopeNote "Land used for high-intensity dry-stock grazing may include rotations for arable or winter forage crops, as well as grazing of non-lactating (dry) dairy cattle, beef cattle, sheep, and cattle." ;
118 skos:usageNote "Stock density is an ideal attribute to assist in classification of intensive vs extensive dry stock grazing." ;
119 skos:changeNote "A prior definition of this classes treated deer, goats, and horses as extensive by definition; it should be noted if known." ;
120 skos:changeNote "<https://github.com/manaakiwhenua/nzsluc/issues/66>" ;
121 .
122
123 luc:extensive_dry_stock a skos:Concept ;
124 skos:inScheme lus:nzslum0.4.0 ;
125 skos:broader luc:grazing_modified_pasture_systems ;
126 skos:notation "2.2.3" ;
127 lu:colour "#ABA248" ;
128 skos:prefLabel "Extensive dry stock"@en ;
129 skos:definition "Grazing on modified pastures with relatively fewer inputs, lower likelihood of irrigation, and lower stocking rates; more likely on hill, hard-hill, or high-country." ;
130 skos:scopeNote "Grazing livestock other than dairy, sheep or beef should usually be captured in this class (though sheep or beef can be included where there is a high proportion of indigenous vegetation for grazing land, land use should be classified as grazing native vegetation)." ;
131 skos:scopeNote "Where there is a high proportion of indigenous vegetation for grazing land, land use should be classified as grazing native vegetation." ;
132 skos:scopeNote "Arable or winter forage crops are unlikely to be common rotations in this land-use category." ;
133 skos:usageNote "The specific commodity or commodities should be noted if known." ;
134 .
135
```

HTML (+ PDF, Markdown, XLSX, ...) –
built, readable, aligned

grazing modified pasture systems #357735

Land used for high-intensity dry-stock grazing may include rotations for arable or winter forage crops, as well as grazing of non-lactating (dry) dairy cattle, beef cattle, sheep, and cattle.

Land under pasture at the time of mapping may be in a rotation system, so that at another time the same area may be, for example, under cropping.

Dairy support is land that is used to support non-lactating dairy stock (dry cows, heifers and calves). It will include any feed required, and will often include winter cropping (dependent), along with cereal crops, such as maize, barley, and wheat. It can also include feed that is cut and carried to the milking platform. Dairy support land should be classified as an arable use.

The ability to distinguish between dairy and dry-stock production is provided by including the relevant commodity information, such as 'cattle dairy', 'cattle beef', 'sheep', etc. This should be recorded if appropriate. Crops used in rotation should also be recorded as commodities, if known.

Intensive dry stock #807735

Land on which milking cows (or other stock, such as goats or sheep) are grazed during the milking season, where dairy is the primary purpose of the land.

Dairy production systems can include rotations of grazed forage crops and maize for dry-stock grazing, but this class should only be used where dairy is the primary purpose of the land.

Where the land is permanently used for dry-stock grazing, it should be classified under dry-stock grazing.

Extensive dry stock #357735

Land used for high-intensity dry-stock grazing may include rotations for arable or winter forage crops, as well as grazing of non-lactating (dry) dairy cattle, beef cattle, sheep, and cattle.

Land used for high-intensity dry-stock grazing may include rotations for arable or winter forage crops, as well as grazing of non-lactating (dry) dairy cattle, beef cattle, sheep, and cattle.

Stock density is an ideal attribute to assist in classification of intensive vs extensive dry stock grazing. (i.e. flat/hill/high country) can be used as a proxy where stock density data is not available.

A prior definition of this classes treated deer, goats, and horses as extensive by definition; it should be noted if such systems can also be intensive.

https://github.com/manaakiwhenua/nzsluc/issues/66

2.2.3 Extensive dry stock #ABA248

Grazing on modified pastures with relatively fewer inputs, lower likelihood of irrigation, and lower stocking rates; more likely on hill, hard-hill, or high-country.

Scope note Grazing livestock other than dairy, sheep or beef should usually be captured in this class (though it does not exclude sheep or beef).

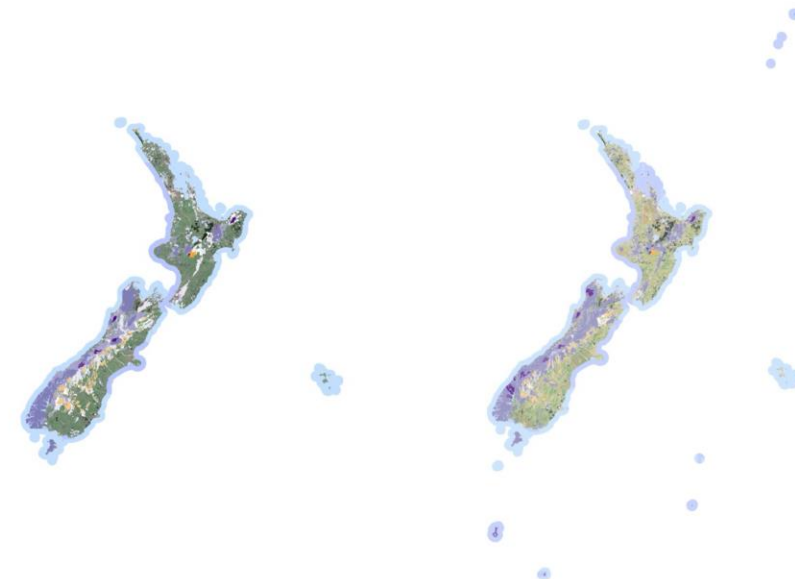
Scope note Where there is a high proportion of indigenous vegetation for grazing land, land use should be classified as grazing native vegetation.

Scope note Arable or winter forage crops are unlikely to be common rotations in this land-use category.

Usage note The specific commodity or commodities should be noted if known.

Part 2 - What the 2026 map does differently

- **Tertiary detail:** 2025 mapped 8 classes at tertiary level; 2026 maps **18**
 - 2.2.1 Dairy · 2.2.2 Intensive dry stock · 2.2.3 Extensive dry stock
 - 3.1.1 High-, 3.1.2 Medium-, 3.1.3 Low-density, 3.1.4 Rural residential
 - 3.2.1 Outdoor recreation · 3.2.2 Indoor recreation · 3.2.3 Community services
- **New secondary class** 2.9.0 Smallholder farm
- **Produced and validated per region** (19 regional subsets, one LRIS set of 18 items) rather than one national run: regional error profiles, regional revisions without a national rebuild
- **Coverage:** offshore islands (Chathams, Kermadecs, subantarctic) now included; PAN-NZ 2026 inherited
- **Inputs:** LCDB v6, LINZ Topo50 additions, Stats NZ urban/rural 2026, consents from 11 councils



2.2.0 Grazing modified pasture → three tertiary classes



2.2.1 Dairy

- Milking platform and dairy support land, incl. forage/maize rotations for the operation
- Evidence, strongest first: dairy effluent discharge consent → pastoral consent naming a dairy commodity → NDVR actual use 11 / category D → LUM cover 502 (dated)
- Effluent consent or irrigation-with-effluent also tags ancillary use 2.7.3
- Now shares the slope/LUC modifiers with 2.2.2: flat, high-capability land corroborates; steep or low-capability land penalises

2.2.2 Intensive dry stock

- Common pastoral evidence pool: NDVR finishing/store codes, pastoral consents, LUM grazed grassland, Crown pastoral lease
- Distinguished by **NZLRI**: slope A–D and LUC 1–5 → intensive (beef/lamb finishing, store stock, non-lactating dairy support)
- Contradiction penalty when LUM says 502 dairy or a consent commodity includes “cattle dairy”

2.2.3 Extensive dry stock

- Slope E–H and LUC 6–7 → extensive hill, hard-hill and high-country pasture
- Default where NZLRI is absent (Chathams, Stewart Island, offshore): flat-terrain evidence is required to be intensive
- Refinements: irrigation split into its own modifier; LUM natural/planted forest now makes 2.2.3 impossible; neutral LCDB covers narrowed to 41, 44, 55
- Terrain is a proxy only because farm-scale stock density is unavailable

3.1.0 Residential → four tiers by density and setting



Urban

- **3.1.1 High-density:** NDVR actual use 92 (multi-unit), 93, 95, 96, category RA; improvements text: apartment, part-floor, boarding house, large unit counts. Ruled out by over-size, industrial zoning, single-unit indicators
- **3.1.2 Medium-density:** 92 without high-density indicators, 95, 96, 22; text: town house, terrace, flat, rest home. 93 motels/holiday parks and 94 hotels route to 3.3.0 Commercial
- **3.1.3 Low-density:** 91 single unit confirmed against Topo50 residential areas, 97 bach, category R; text: dwelling, pool, sleep-out, double garage
- Kāinga Ora properties are absent from the DVR → LINZ Central Record of State Land (CRoSL) fills the gap for 3.1.2/3.1.3

Peri-urban (boundary with class 2)

- **3.1.4 Rural residential:** 21/22 lifestyle with a confirmed dwelling, 97 in a rural area, category L; only 0.4–2 ha
- > 2 ha residential/lifestyle → **2.9.0 Smallholder farm** (Class 2), or another Class 2 use with production evidence
- Over-sized urban properties point toward 3.1.4; high/medium indicators and industrial zoning push out of 3.1.3/3.1.4

3.2.0 Public recreation and services → three tertiary classes



3.2.1 Outdoor recreation

- Parks, trails, sportsgrounds, camping grounds, zoos, botanic gardens, recreational reserves, MTB parks
- Evidence: Topo50 sports facilities and golf courses; HAIL C2 (gun clubs, ranges); PAN-NZ 2026 racecourse reserves with LCDB Urban Parkland; recreational reserves without indigenous cover; NDVR outdoor recreation
- **Explicit boundary with Class 1:** reserves with significant native bush or protected status → 1.1.7; 3.2.1 is roughly its complement

3.2.2 Indoor recreation

- Sports centres, gyms, pools, indoor arenas
- Identified from NDVR improvements descriptions only → a thin class (n = 38 in validation)
- Boundary with 3.2.1 also explicit

3.2.3 Community services

- Education, public health, libraries, museums, courts, prisons, civic buildings, emergency services, marae, religious buildings, cemeteries
- Topo50 cemeteries and urupā; LINZ national facilities (hospitals, schools); NDVR improvements list (fire stations, churches, monuments...)
- Motivation for the split: PAN-NZ 2026 exposed confusion over whether local-purpose reserves are conservation or recreation

Input data revisions for 2026



- **LCDB v6** (incl. Chathams) replaces v5: land cover moves from nominal 2018 to 2023. Used as the contradiction check in 2.2.x, the Urban Parkland signal in 3.2.1, part of the 1.3.6 exclusion, and more
- **NZLRI slope and LUC** now used directly as physiographic evidence for 2.2.1–2.2.3 (flat/LUC 1–3 corroborates dairy; steep/LUC 7–8 the opposite)
- **LINZ Topo50**: beacons and masts → 3.5.0 Utilities; racetracks (with sport noted) → equestrian/horse-training operations (raised by WRC)
- **Stats NZ urban/rural 2026** boundaries
- **PAN-NZ 2026**: more than a temporal update: wider coverage, omission/commission fixes, and a re-mapped legislation → IUCN crosswalk (Hughes 2025) → Class 1.1.x changed considerably
 - Non-IUCN and unmapped reserves sit with IUCN VI under **1.1.7**, except SNAs, Ramsar and World Heritage sites, Crown pastoral leases (→ 2.2.3) and racecourse reserves (→ 3.2.1)
- Not yet used: annual 10 m Sentinel-2 land cover (recency and resolution vs LCDB's thematic depth); LiDAR products

Regional council consents: the big new evidence source



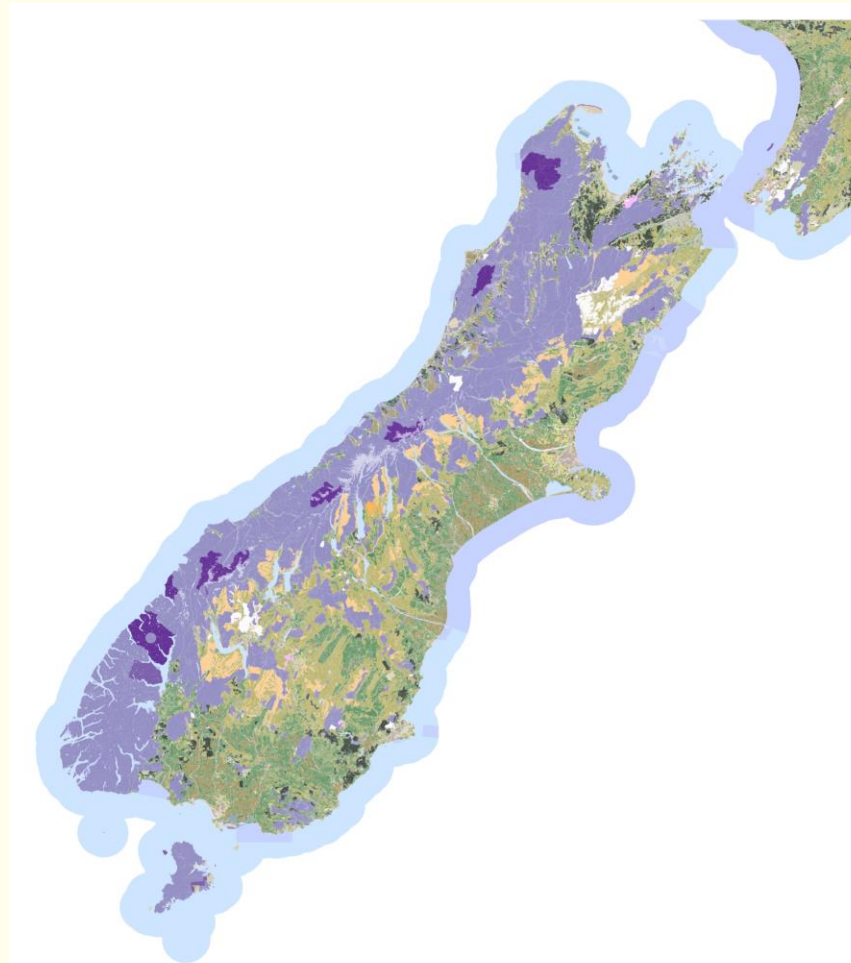
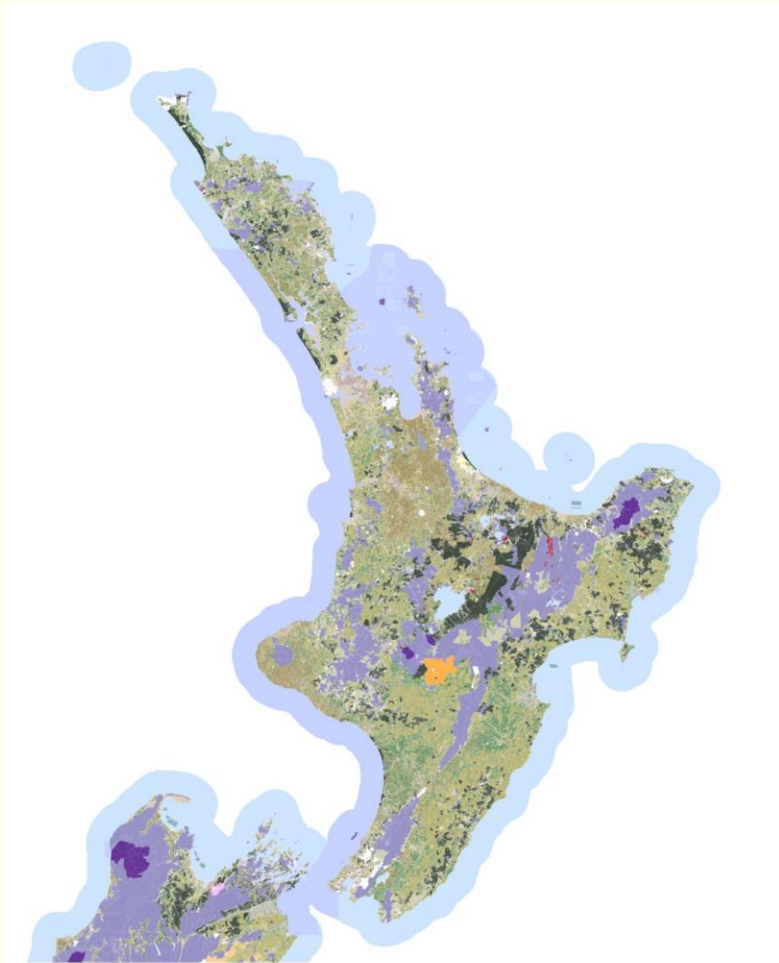
Now included (11 councils)

- **Bay of Plenty** dairy consents → effluent evidence
- **Canterbury** active consents: irrigation, dairy effluent discharge
- **Southland, Hawke's Bay** (pre-existing; expanded use)
- **Greater Wellington** dairy shed effluent; plantation forestry with species
- **Otago** dairy farm and intensive winter grazing consents
- **Taranaki** afforestation; animal-waste discharge
- **Waikato** dairy, general pastoral, forestry
- **Marlborough** dairy/piggery discharge (→ 2.2.1, 2.6.0)
- **West Coast** dairy and forestry; **Horizons** point companion layer

How a point becomes land-use evidence

- Most consents are **points** (a discharge pipe, an effluent pond, an address), not farm polygons
- Pipeline asks “which property is this on?” and lets the whole property carry the evidence → a shed consent supports Dairy across the milking platform and run-off
- Risks: very large, partly leased or multi-use properties; misplaced discharge points → point evidence gets **lower confidence** than polygons (corroboration, not observation)
- Polygon sources (e.g. some sources in Marlborough, West Coast, Wellington) map directly at higher confidence
- Ingested with **Kart** at date-pinned commits; weekly changesets accumulate so future rules can be replayed back in time

NZLUM v0.4 2026

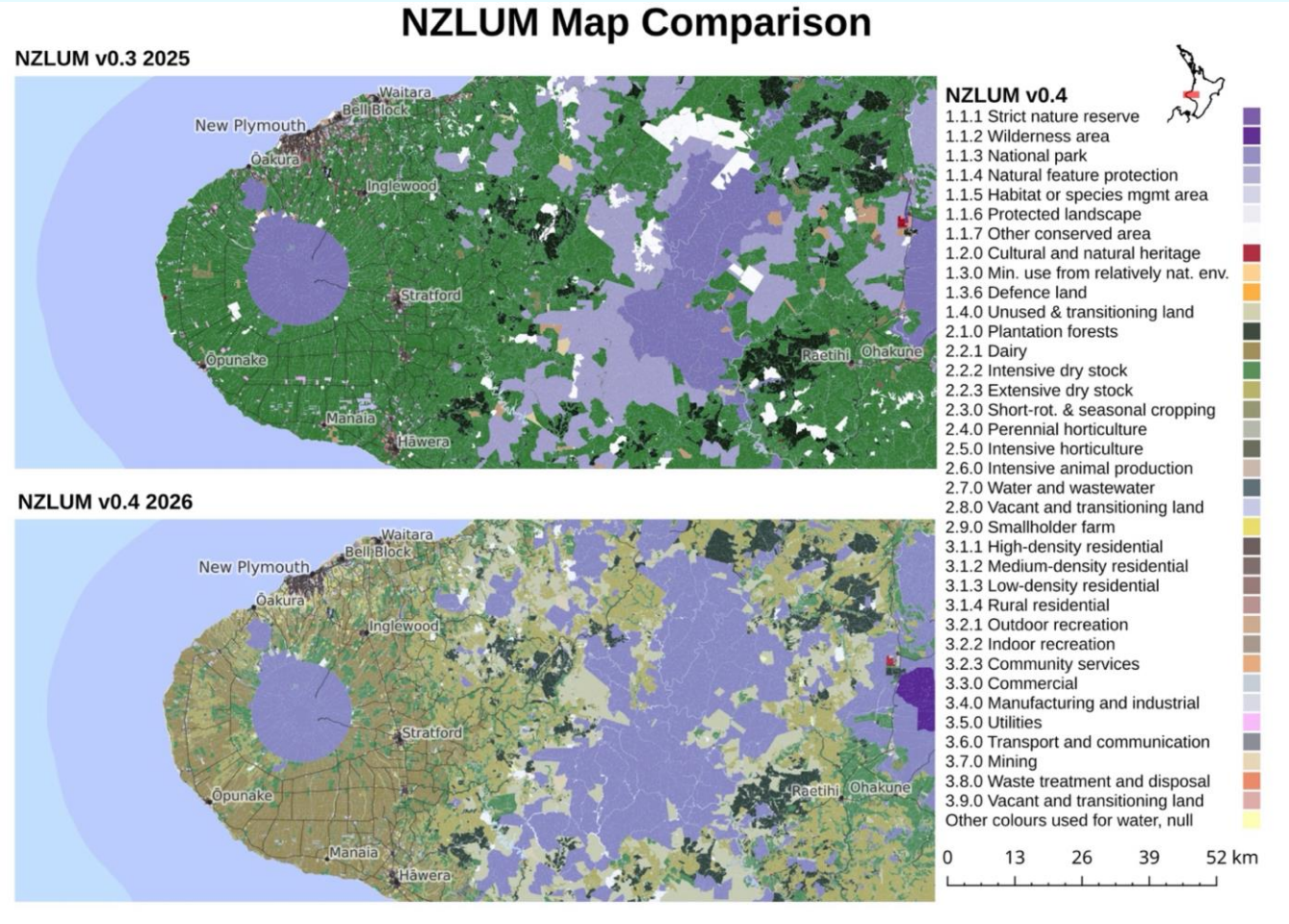


NZLUM v0.4 2026

- 1.1.1 Strict nature reserve
- 1.1.2 Wilderness area
- 1.1.3 National park
- 1.1.4 Natural feature protection
- 1.1.5 Habitat or species management area
- 1.1.6 Protected landscape
- 1.1.7 Other conserved area
- 1.2.0 Cultural and natural heritage
- 1.3.0 Minimal use from relatively natural environments
- 1.3.6 Defence land
- 1.4.0 Unused and transitioning land
- 2.1.0 Plantation forests
- 2.2.0 Grazing modified pasture systems
- 2.2.1 Dairy
- 2.2.2 Intensive dry stock
- 2.2.3 Extensive dry stock
- 2.3.0 Short-rotation and seasonal cropping
- 2.4.0 Perennial horticulture
- 2.5.0 Intensive horticulture
- 2.6.0 Intensive animal production
- 2.7.0 Water and wastewater
- 2.8.0 Vacant and transitioning land
- 2.9.0 Smallholder farm
- 3.1.0 Residential
- 3.1.1 High-density residential
- 3.1.2 Medium-density residential
- 3.1.3 Low-density residential
- 3.1.4 Rural residential
- 3.2.0 Public recreation and services
- 3.2.1 Outdoor recreation
- 3.2.2 Indoor recreation
- 3.2.3 Community services
- 3.3.0 Commercial
- 3.4.0 Manufacturing and industrial
- 3.5.0 Utilities
- 3.6.0 Transport and communication
- 3.7.0 Mining
- 3.8.0 Waste treatment and disposal
- 3.9.0 Vacant and transitioning land
- Additional colours used for forms of water, and null

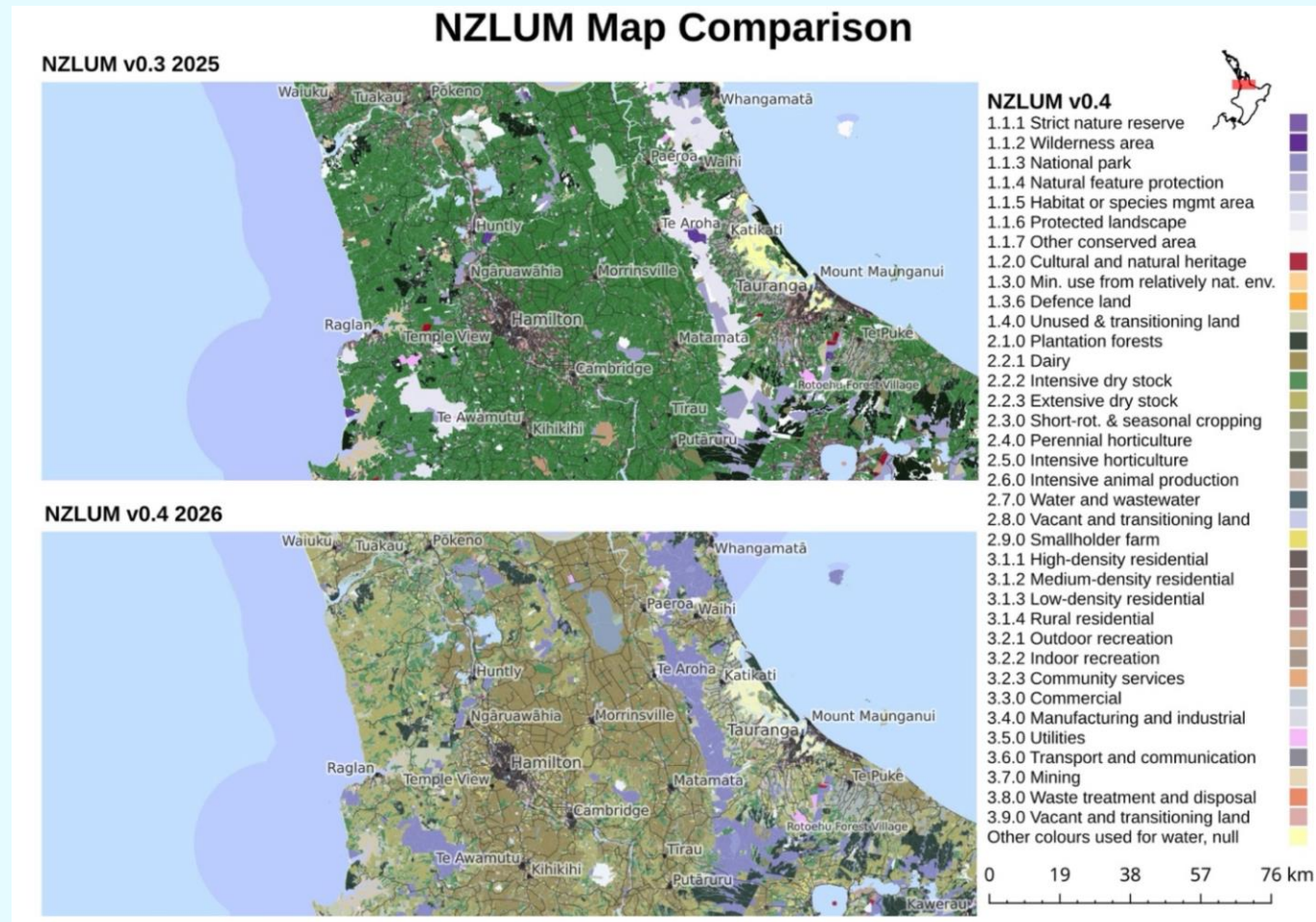
Class only; the data also carry commodity, management practice, tenure, zoning, land cover, water and confidence.

Before / after: Taranaki



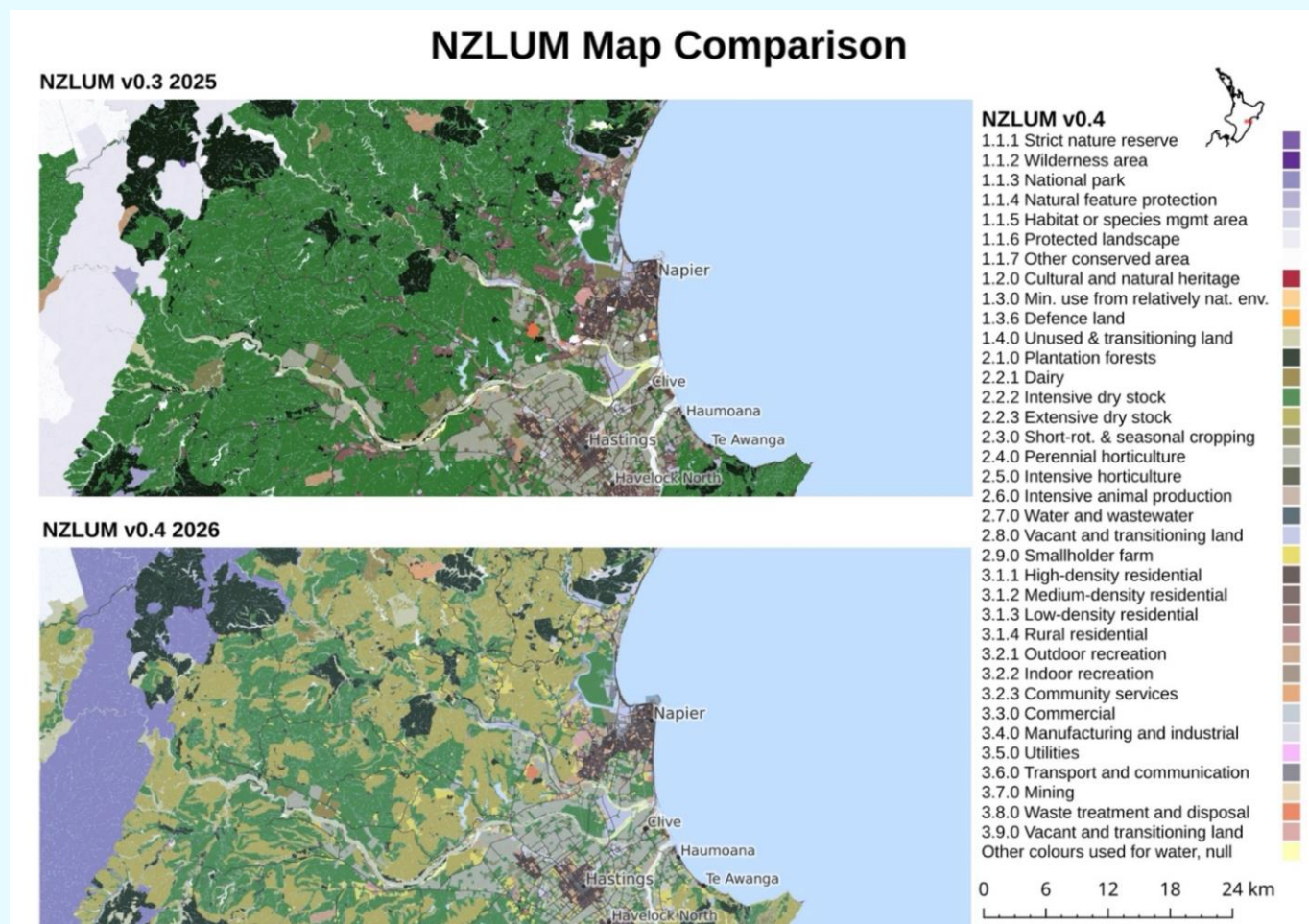
Top: NZLUM v0.3 2025. Bottom: NZLUM v0.4 2026. Report Figure 4.

Before / after: Waikato



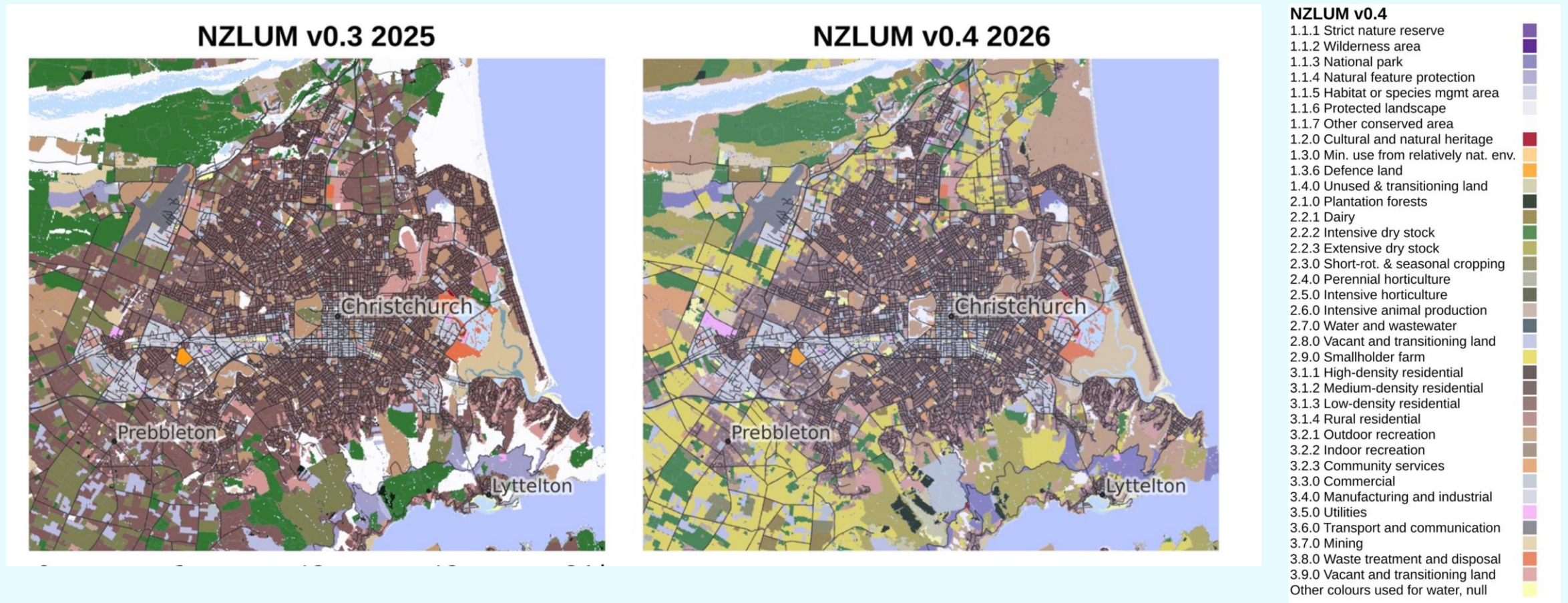
Top: NZLUM v0.3 2025. Bottom: NZLUM v0.4 2026. Report Figure 8.

Before / after: Hawke's Bay



Top: NZLUM v0.3 2025. Bottom: NZLUM v0.4 2026. Report Figure 7.

Before / after: Christchurch and the new Class 3 detail



Left: NZLUM v0.3 2025. Right: NZLUM v0.4 2026. Report Figure 10.

Part 3 - Validation: sampling design

- Stratified random samples drawn **independently for 19 regional subsets** (2025: once nationally)
- Sample size per stratum from the standard proportion formula: 95% confidence, **25% margin of error**, finite-population correction, floor of 10 per stratum
- **Area weighting:** $\log(1 + \text{area})$: a polygon 100× larger gets $\approx 7\times$ the weight, so neither giant polygons nor the mass of tiny ones dominate
- **7,763** polygons validated by remote desktop inspection of aerial imagery; two validators (13 and 6 regions); **no redundant validation**
- Pooled across regions the margin of error approaches $25\% / \sqrt{19} \approx \mathbf{5.7\%}$ for widespread classes; a class present in one region stays at 25%
- Outcomes recorded: true-tertiary, true-secondary, maybe, false
- Tools released under GPL: **spatial-stratified-sampling** (CLI) and **check-a-kea** (QGIS plugin), github.com/manaakiwhenua



Headline accuracy

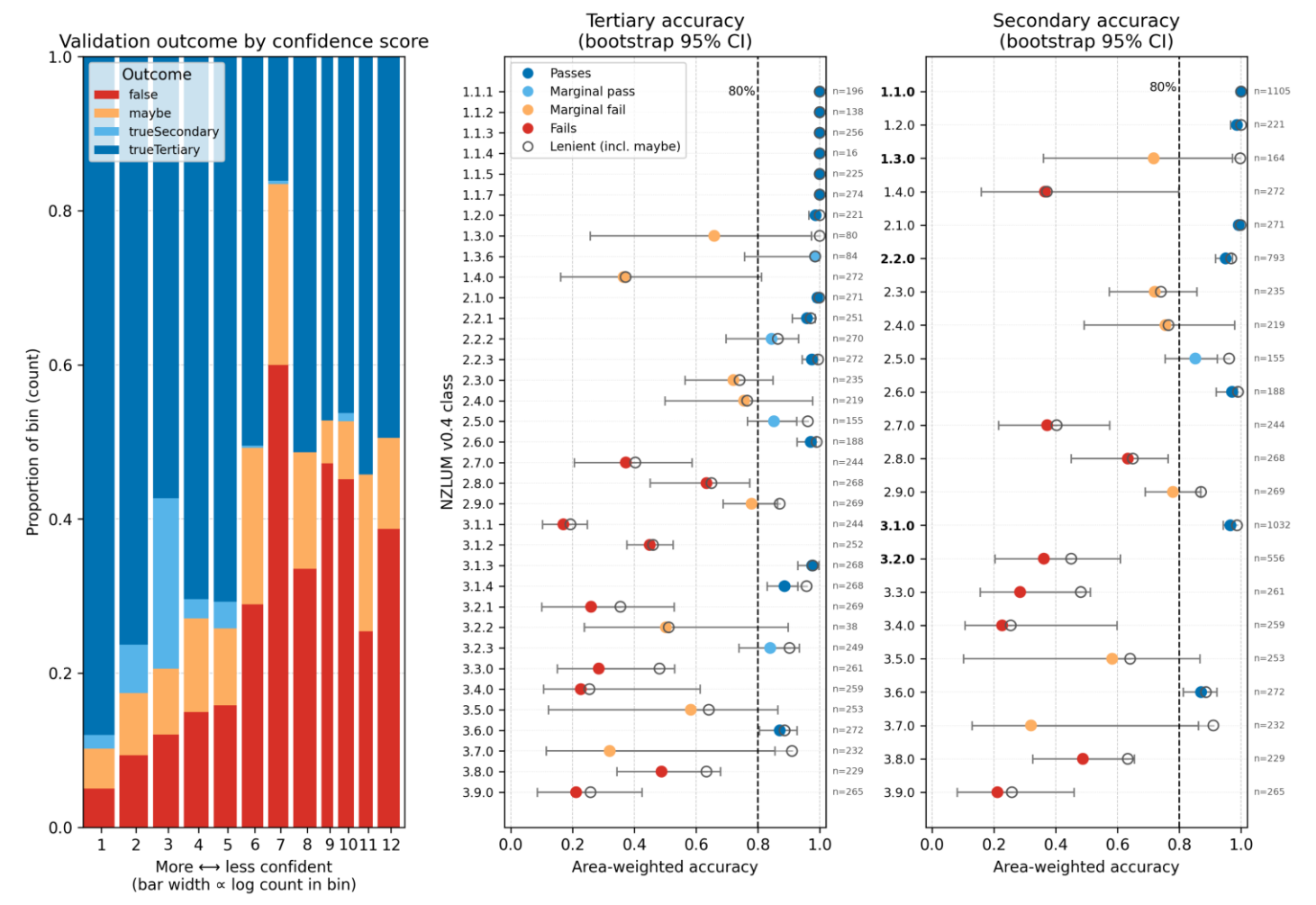


Overall (all classes)	Area-weighted tertiary	Area-weighted secondary	Count-based tertiary	Count-based secondary
Strict	98.9%	99.4%	71.7%	75.9%
Lenient (maybe = correct)	98.9%	99.4%	75.9%	85.1%

By primary class (strict tertiary)	n	Count-based strict	Count-based lenient	Area-weighted strict	Area-weighted lenient
1 Conservation and minimal use	1,762	85.9%	94.2%	99.4%	99.8%
2 Production agriculture and plantations	2,642	73.2%	81.8%	93.5%	95.2%
3 Built environment	3,359	61.8%	72.0%	42.8%	59.2%

- Area weighting flattens Classes 1 and 2: big polygons (national parks, uniform grazing) are easy. Sample was drawn with a muted area bias, so weighting is defensible but not neutral
- Class 3 **inverts**: area-weighted 42.8% (95% CI 26.0–62.4%) is below count-based 61.8% → its large polygons are disproportionately wrong (3.2.1 outdoor recreation, 3.5.0 utilities)

Per-class accuracy (report Figure 11)



The good, the bad, and the ugly



Passes (CI above 80%)

- All of **1.1.x** nature conservation; **1.2.0**; **1.3.6** defence
- **2.1.0** plantation forests
- **2.2.1 Dairy** and **2.2.3 Extensive dry stock**, both new this year
- **2.6.0** intensive animal production
- **3.1.3** low-density and **3.1.4** rural residential (new)
- **3.6.0** transportation

Marginal (80% inside the CI)

- **2.2.2 Intensive dry stock**: passes marginally; needs more than terrain to match 2.2.1/2.2.3
- **2.3.0 cropping** and **2.4.0 perennial horticulture**: fail marginally; important classes needing more information
- **2.5.0**, **3.2.3** community services: marginal passes
- **1.3.0**, **2.9.0** smallholder farm, **3.2.2**, **3.5.0** utilities, **3.7.0** mining: marginal fails

Fails (CI below 80%)

- **3.1.1 High-density residential** < 20% and **3.1.2 Medium-density** ≈ 45%: wrong tier, right secondary class
- **3.2.1 Outdoor recreation** 25%, **3.5.0 Utilities** 39% (secondary), **3.9.0** vacant/transitioning 44%
- **1.4.0**, **2.7.0** water & wastewater, **2.8.0**
- **3.3.0** commercial, **3.4.0** industrial, **3.8.0** waste
- Validation of “minimal use” and transition classes is itself unreliable from imagery

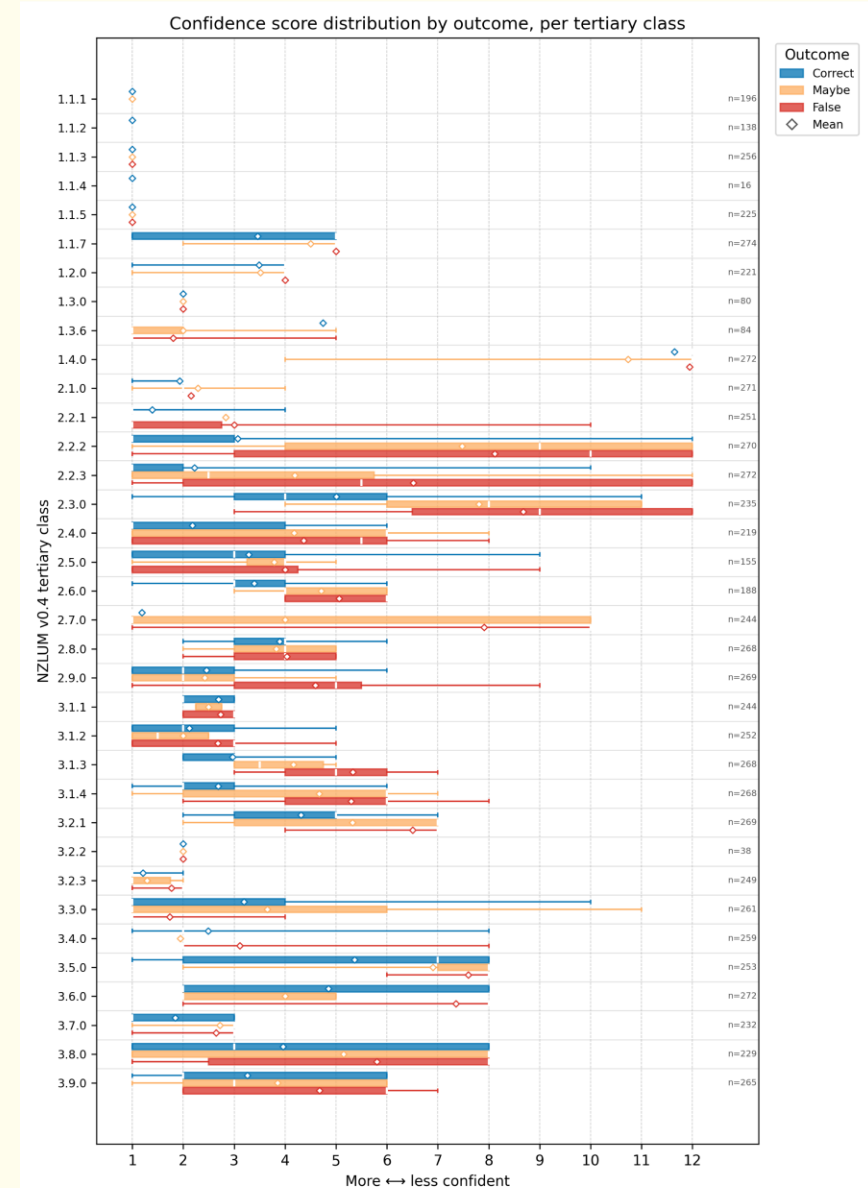
Class 3: right secondary class, wrong tertiary class



- **3.1.1 High-density residential:** 0% false, but only 12% true at tertiary level: consistently Residential, rarely correctly high-density
- **3.1.2 Medium-density:** 52% true at tertiary level, 98% at secondary; low false rate
 - density rules need a higher evidential threshold or new data: **LiDAR building footprints with height** would be an excellent indicator, joined to properties
- **3.2.1 Outdoor recreation** and **3.5.0 Utilities:** disproportionately large natural spaces: peri-urban parks that are not really urban, reservoir catchments that are not really utility-focused: many arguably Class 1
 - **3.2.1** depends on Stats NZ administrative urban/rural definitions, whose generous peri-urban zones drive apparent misclassification (→ recommendation 7)
 - **3.5.0** definition question for this group: are natural water collection areas utilities? (→ recommendation 8)
- **3.4.0 Industrial** validated poorly despite being visually prominent: we lack inputs derived from imagery segmentation (→ suggested change)

Is the confidence score trustworthy?

- NZLUM confidence: 1 (most) to 12 (least confident), set by the classifier, not the validator
- **Bin 1** (n = 2,738; largest by area and count): **88.0%** correct by count at tertiary level
- Bins 2–5: 76.3%, 57.3%, 70.4%, 70.7% (informative, not monotonic)
- Bin 6: 50.5%. **Bin 7 (n = 230) is an outlier: 16.1% correct, 60.0% false**, worse than the less confident bins
- Bins 8–12: noisy 46–54% correct, 25–47% false; bins 7–12 hold 1,265 polygons = 16.3% of the sample
- “Maybe” = 9.2% overall, rising from 5–8% (bins 1–3) to 12–24% (bins 4–7): low-confidence calls are ambiguous even to a human
- Right: confidence distribution of correct / maybe / false polygons per class (report Figure 12)



Usefully uncertain vs confidently wrong; limits of the validation



Reading Figure 12

- **Good pattern:** false calls at low confidence, e.g. 2.3.0 cropping (false mostly at 6–12), 2.2.2/2.2.3 dry stock: the score predicts reliability
- **Bad pattern:** false calls at high confidence, e.g. **3.7.0 Mining** (false at 1–3), 3.4.0 industrial, 3.2.3, 1.1.7: the score cannot be trusted there
- Recommendation 1 follows: prioritise data for classes that validate poorly **and** are inversely calibrated

What validation cannot tell us

- Strict → lenient gap for **1.3.0** minimal use: 73% → 93%: “minimal use” is hard to judge from imagery
- Same for all vacant/transition classes (1.4.0, 2.8.0, 3.9.0): desktop validation is insufficient
- Operator error is unquantified; no redundant validation; two validators
- Count vs area weighting can diverge sharply for a class dominated by a few large, easy polygons
- Ideal: **multiple redundant validators validating places they know**
- Idea: independent area totals as a cross-check

Changes to the LUIS classifier software



- **Compressed DGGS representation:** inputs stored so uniform areas (the interior of a national park) take few rows, with no loss of information
 - Large reduction in compute and disk → a national map at a **higher H3 resolution is now feasible** (better for roads, rivers and Class 3)
- **Kart** (open-source geospatial version control, Koordinates) supported as an input source: LRIS tracks the council consent ArcGIS REST endpoints weekly and commits changes automatically
 - Classifier can pin any input to a snapshot date → future rule changes can be **replayed back in time**, separating real land-use change from method change (Meeting 1 item I)
 - Clones are integrity-checked; REST downloads were not (we saw lost packets / random holes in 2025)
- Only consents (and LINZ topo) are tracked this way so far; recommend extending to all vector inputs
- All 2026 configuration changes are public: github.com/manaakiwhenua/luis-config-nzlum

Part 4 - Report recommendations: data and classification (1–4)



1. **Validation-led data acquisition.** Start with the classes that validate worst; most urgent where validation performance is inversely related to stated confidence
2. **Bring in raster inputs** (10 m Sentinel-2 land cover, LiDAR canopy height). Straightforward to integrate; use first as **exclusions** for poor performers such as 3.5.0 Utilities
3. **Councils: encourage to publish more data at all**, especially asset registers (even point locations) for utilities, possibly by working with utilities companies? Open standards remain desirable but secondary; regionally inconsistent consents are used because no standard exists.
4. **Standalone crop and perennial-horticulture maps** (e.g. deep learning) produced independently, then integrated: lift 2.3.0 and 2.4.0 above 80% and enable tertiary mapping
 - 2.4.1 tree crops, 2.4.2 vine crops, 2.4.3 other perennial; 2.3.1 arable, 2.3.2 arable + livestock, 2.3.3 short-rotation horticulture, 2.3.4 flowers/bulbs/turf: high interest

Report recommendations: resolution, products and rules (5–8)



5. **Resolution.** H3 resolution 11 is insufficient for the map to be useful in detail. Map nationally at **resolution 12**, and urban areas separately at **resolution 13**; the 2026 software improvements make this feasible
 6. **A property-boundary version** of the map should be put before NZLUSG: more useful for council staff and others working at property scale
 7. **3.2.1 Outdoor recreation** rule: better test whether an area is truly urban or peri-urban rather than relying on Stats NZ urban/rural boundaries
 8. **3.5.0 Utilities** definition to be revised by NZLUSG, e.g. whether natural water collection areas belong there
- **Pilot status (S7.2):** not all tertiary classes reach 80% → keep publishing with a **‘pilot’ disclaimer**; reconvene NZLUSG (today) to consider the map, postulate further (progressively minor) changes and **prioritise tertiary classes** for the next iteration

Carried over from Meeting 1 (open or proposed, not decided)



Classification

- **Defence in built-up contexts** (Linton, Ōhakea): add a Class 3 defence class with subclasses per service branch (proposed)
- **3.2.3 Community services** as its own secondary class rather than under recreation? (raised by LINZ; no decision)
- **2.9.0 tertiary size gradation** following land-fragmentation classes (decision D: “at a future point”)
- **Nature conservation vs zoning**: revisit when a zoning standard is operative

Data

- **NAIT** access for farm-scale stocking rates (ministerial lobbying suggested); APS data as an alternative
- State data-source caveats (e.g. DVR) more clearly

Process & standards

- **ISO 19144-3:2024 LUML**: describe NZLUM formally? Talk to ABARES (action G, proposed)
- **Real change vs method change** in future releases (item I, open): Kart replay is a partial answer
- **Provenance**: keep current approach; PROV-N / lookup noted for later (H)
- **Engagement**: survey got zero responses → use LRIS statistics to find top users per domain and call them (N)
- **Funding**: find a ‘beachhead’ end-user able to tolerate a prototype

Suggested changes so far (unagreed): method and data



- **Stocking rate as an attribute.** Spatially distribute Agricultural Production Census stocking data across the areas mapped in the corresponding agricultural classes → estimated stocking rate per area in the attribute table
- **Direct raster inputs**, used to eliminate impossible class assignments or to adjust the confidence of the assigned class:
 - NDVI
 - Sentinel-2 land cover at 10 m (LRIS layer 123984, “NZ cost-effective land cover 2023-24”)
- **Automated segmentation for industrial land.** 3.4.0 validated very poorly although it is visually prominent; the missing piece is input derived from segmenting aerial imagery. Combined with zoning and existing sources → a high-confidence, well-validating class
 - If efficient and successful, extend the approach to other classes chosen by: poor 2026 validation + adequate training data + an obvious visual pattern a machine can learn

Suggested changes so far (unagreed): output format and validation



Output format

- **Aggregate DGGS cells to land parcels / property boundaries:** classify the whole polygon by the dominant land use by area; record the next-highest as a secondary land use; possibly retain all relevant commodities
- **Increase resolution:** rural areas at H3 resolution 12 (+1), urban areas at 13 (+2)
 - Open question: define “urban” from NZLUM classes themselves, or from a Stats NZ boundary such as functional urban areas?

Validation

- **Compare mapped areas** of agriculture, plantation forestry and cropland with independent totals: APS data, sector reporting
- From the report: redundant validators; local knowledge; recognise that minimal-use and transition classes cannot be validated from imagery alone
- Keep the 80% tertiary threshold as the graduation criterion from pilot status?

What we are asking NZLUSG today: suggest, agree, reject or rank



Classification system

- Revise **3.5.0 Utilities** definition (water collection areas?)
- Revise **3.2.1** (outdoor recreation) urban / peri-urban criterion
- **Defence** under Class 3; **3.2.3** placement
- **2.9.0** (smallholder farm) tertiary size gradation
- Any other class, definition or attribute change from your needs
- Scale of the next release: **v0.4.x** (compatible) or **v0.5**?

Method, data and output

- Rank:
 - APC stocking-rate attribute
 - NDVI / Sentinel-2 rasters
 - imagery segmentation (industrial first)
 - crop & horticulture maps
 - LiDAR building heights for residential
- To pursue: council utility asset registers
- **Parcel-aggregated product:** yes / no / alongside the grid?
- Resolution 12 rural / 13 urban, and how “urban” is defined
- Keep the **pilot** label until every tertiary class $\geq 80\%$

Tertiary classes and direction

- Which tertiary classes next: 2.3.x / 2.4.x (needs crop maps), 3.4.x (segmentation), 3.5.x, 3.3.x, 2.9.x, others?
- Is NZLUM v0.4 2026 adequate for your organisation's uses? Where does it fall short?
- Direction: less change to the system, more to map quality and precision: agreed?
- Validation
 - Area totals vs APS / sector data
 - Who could contribute local validators? We'd provide data, tools and training

In summary

- **v0.4** published as a versioned SKOS vocabulary with change notes; peri-urban gap closed with 3.1.4 and 2.9.0
- **2026 map:** 18 tertiary classes, regional production, 11 councils' consents, LCDB v6, PAN-NZ 2026, offshore islands
- **Validation:** Classes 1 and 2 strong by area (99% / 94%); new 2.2.1 and 2.2.3 pass; Class 3 tertiary weak (43% by area) and confidence miscalibrated for some classes
- **Recommendations + suggested changes + your priorities** decide the next map; the pilot label stays until tertiary classes clear 80%
- Available now: LRIS set 7516 (view, API, download); everything else at <https://nzlum.landcareresearch.co.nz>

The screenshot displays the LRIS Portal interface for the NZLUM v0.4 2026 dataset. The header shows the LRIS Portal logo and navigation options. The main content area features a map of New Zealand with a red dot indicating the location of the dataset. Below the map, there is a table with metadata for the dataset, including Date Added, Last Updated, Views, and Set ID. A detailed description of the dataset is provided, stating it is a land-use map of New Zealand at a nominal date of June 2026, using the New Zealand Land Use and Management (NZLUM) v0.4 classification system. The page also includes a license section and a category section.

LRISPORTAL Manaaki Whenua Landcare Research More information services

<https://lris.scinfo.org.nz/set/7516-nzlum-v04-2026/> Landcare Research via LRIS Portal

NZLUM v0.4 2026 ☆ ... Map

Info Contents Services and APIs

 Data Type Set, 18 Items

Date Added	Last Updated	Views	Set ID
+ 13 Jun 2026	🕒 19 Jun 2026	👁 17	📁 7516

A land-use map of New Zealand, at a nominal date of June 2026. The classification system used is the New Zealand Land Use and Management (NZLUM) v0.4. More information can be found at nzlum.landcareresearch.co.nz/

License
🔒 Copyright or licenses may apply

LRIS Portal Category
Economy > Resource utilisation > Land use

Tags

landuse@landcareresearch.co.nz

<https://nzlum.landcareresearch.co.nz>

[LRIS: Iris.scinfo.org.nz/set/7516-nzlum-v04-2026](https://iris.scinfo.org.nz/set/7516-nzlum-v04-2026)

[Classifier configuration: github.com/manaakiwhenua/luis-config-nzlum](https://github.com/manaakiwhenua/luis-config-nzlum)

Thank you